

Year 8 Science – Booklet 14: Biology

Photosynthesis — ANSWERS

Quick Test (p.27)

1. Glucose and oxygen.
2. In the leaves (mainly in the chloroplasts of the palisade cell layer).
3. Nitrates, phosphates and potassium.

Section A – Multiple Choice (5 marks)

1. a) during the day
2. c) xylem
3. d) palisade cells
4. d) nitrate
5. d) stunted growth

Section B (7 marks)

1. Match the mineral to its deficiency symptom (3 marks)

Nitrate → poor root growth and purple young leaves
Phosphate → yellow leaves with dead spots
Potassium → stunted growth and yellow older leaves

2. Labelled leaf section (4 marks)

- a). Palisade cells (contain the most chloroplasts).
- b). Stoma (the opening that lets carbon dioxide into the leaf).
- c). The waterproof layer (cuticle).
- d). The leaf vein (contains the xylem and phloem tubes).

Section C (7 marks)

1. Pondweed photosynthesis experiment

- a). Oxygen.
- b). Collect the gas in a test tube and test it with a glowing splint – if it relights, this confirms the gas is oxygen.
- c). Light and carbon dioxide (2 marks).
- d). Glucose.

2. Iodine test on pondweed leaves

- a). Blue-black.
- b). Starch – showing that photosynthesis had taken place, since the glucose made is stored as starch.